NRF Communication - Robot

Aim:

To design and implement a wireless robot control system using two ESP32-C6 modules and nRF24L01 transceivers, enabling directional robot movement via push-button commands.

Apparatus Required:

S. No	Component	Quantity
1.	ESP32-C6 Dev Board	2
2.	L298N Motor Driver	2
3.	12v DC Motors(300RPM) with Wheel	4
4.	Jumper Wires	As needed
5.	USB Type-C Cable	1
6.	Computer with Arduino IDE	1
7.	Robot Chassis	1
8.	12v Battery	1
9.	Push Button	2
10.	nRF24L01 Modules	2

Theory:

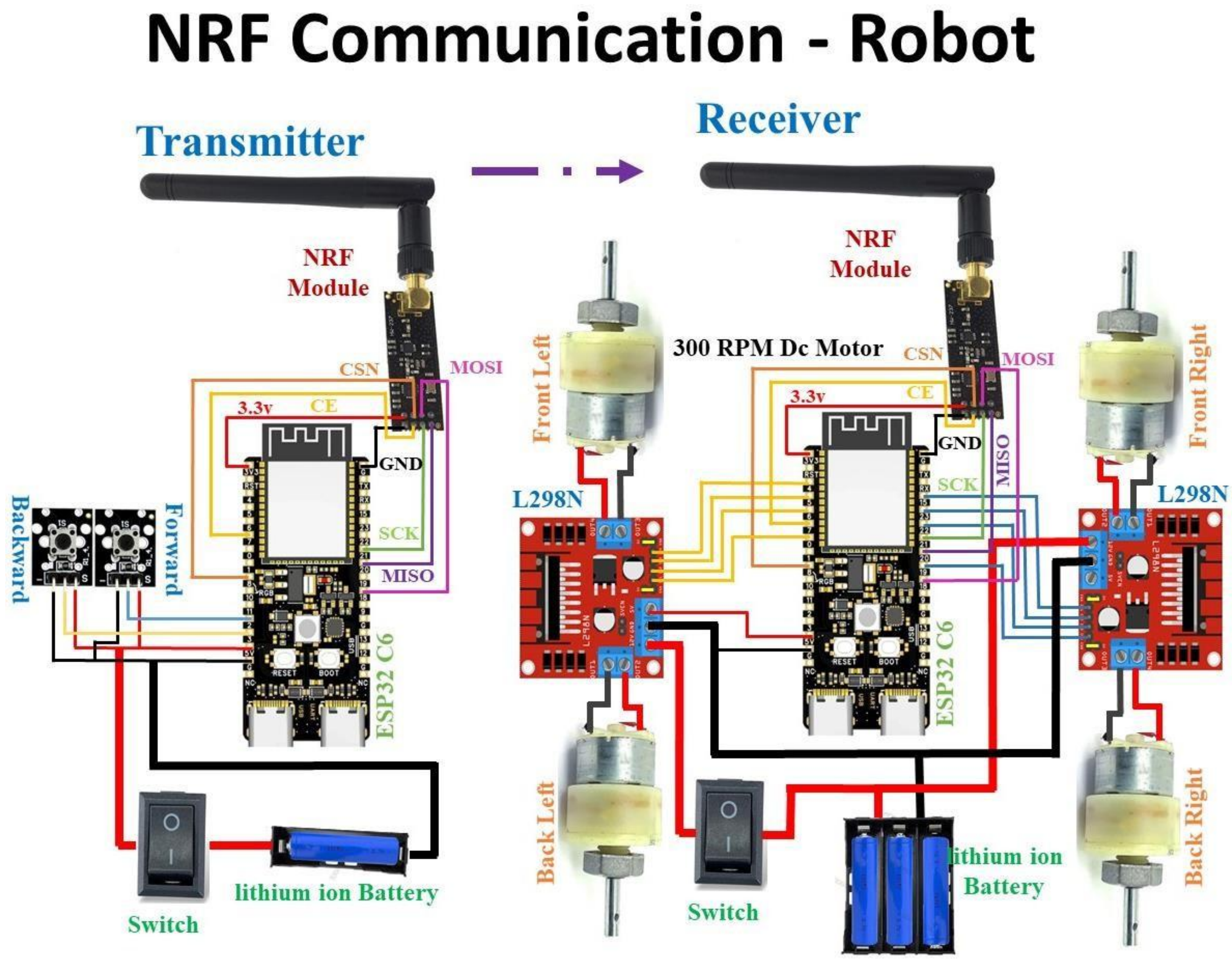
- The nRF24L01 is a low-cost, low-power 2.4GHz wireless transceiver module used for SPI-based communication. In this project:
- One ESP32-C6 with nRF24L01 acts as a transmitter, reading button input and sending control commands (F, B, or S) wirelessly.
- The second ESP32-C6 with nRF24L01 acts as a receiver, interpreting commands and controlling motors using an H-Bridge.
- This enables remote wireless robot control for basic forward and backward motion.

Procedure:

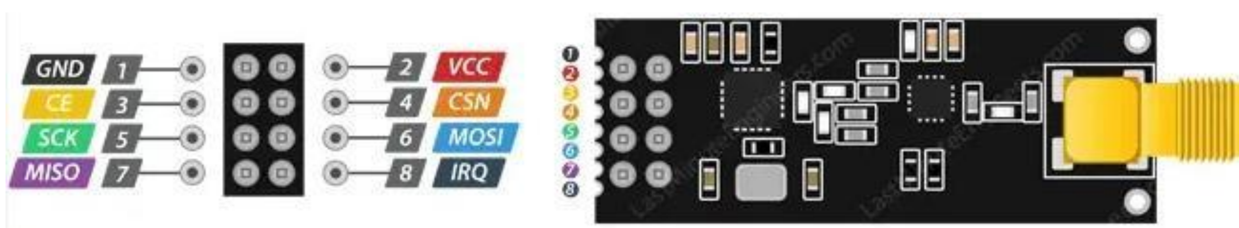
1. Transmitter:
   - Connect the Buttons: Use GPIO2 for forward and GPIO3 for backward. Connect one end of each button to ground and the other to the ESP32 pins with INPUT_PULLUP enabled.
   - Connect nRF24L01 Module: Connect the SPI pins (MISO, MOSI, SCK) to GPIO20, 19, and 21 respectively, and CE to GPIO7, CSN to GPIO8.
   - Upload Transmitter Code: Use Arduino IDE to upload the provided transmitter sketch to the ESP32-C6.
   - Test Serial Output: Open Serial Monitor to verify that the commands ('F', 'B', or 'S') are being transmitted when buttons are pressed.

2. Receiver:
   - Connect Motor Driver Inputs: Connect motor control pins (IN1-IN4) of L298N or other H-Bridge to GPIOs 0, 6, 5, and 4 of ESP32-C6.
   - Connect Motors: Attach DC/Mecanum motors to the outputs of the motor driver.
   - Power the Motor Driver: Use an external 5V–12V battery pack to power the motors through the driver board.
   - Connect nRF24L01 Module: Same as transmitter (SPI + CE & CSN).
   - Upload Receiver Code: Flash the receiver ESP32-C6 with the appropriate sketch.
   - Run and Observe: Press the buttons on the transmitter and observe motor movements on the robot.

Circuit Diagram:



Pinout :



Transmitter Program:

```
#include <SPI.h>
#include <nRF24L01.h>
#include <RF24.h>

#define CE_PIN 7
#define CSN_PIN 8
#define BTN_FWD 2
#define BTN_BWD 3

#define MY_MOSI 19
#define MY_MISO 20
#define MY_SCK  21

RF24 radio(CE_PIN, CSN_PIN);
const byte address[6] = "00001";

char command = 'S'; // 'F' = forward, 'B' = backward, 'S' = stop

void setup() {
  pinMode(BTN_FWD, INPUT_PULLUP);
  pinMode(BTN_BWD, INPUT_PULLUP);
  Serial.begin(115200);

  SPI.begin(MY_SCK, MY_MISO, MY_MOSI, CSN_PIN);
  radio.begin();
  radio.setPALevel(RF24_PA_HIGH);
  radio.setDataRate(RF24_1MBPS);
  radio.setChannel(108);
  radio.openWritingPipe(address);
  radio.stopListening();
}

void loop() {
  if (digitalRead(BTN_FWD) == HIGH) command = 'F';
  else if (digitalRead(BTN_BWD) == HIGH) command = 'B';
  else command = 'S';

  radio.write(&command, sizeof(command));
  Serial.print("Sent: "); Serial.println(command);
  delay(150);
}
```

Receiver Program:

```
#include <SPI.h>
#include <nRF24L01.h>
#include <RF24.h>

#define CE_PIN 7
#define CSN_PIN 8

#define MY_MOSI 19
#define MY_MISO 20
#define MY_SCK  21

// Motor driver pins
#define M1_IN1 0
#define M1_IN2 6
#define M2_IN1 5
#define M2_IN2 4
#define M3_IN1 15
#define M3_IN2 23
#define M4_IN1 22
#define M4_IN2 19

RF24 radio(CE_PIN, CSN_PIN);
const byte address[6] = "00001";

char command;

void setup() {
  Serial.begin(115200);

  // Set motor pins as outputs
  pinMode(M1_IN1, OUTPUT); pinMode(M1_IN2, OUTPUT);
  pinMode(M2_IN1, OUTPUT); pinMode(M2_IN2, OUTPUT);
  pinMode(M3_IN1, OUTPUT); pinMode(M3_IN2, OUTPUT);
  pinMode(M4_IN1, OUTPUT); pinMode(M4_IN2, OUTPUT);

  // Start SPI with custom pins
  SPI.begin(MY_SCK, MY_MISO, MY_MOSI, CSN_PIN);

  radio.begin();
  radio.setPALevel(RF24_PA_HIGH);
  radio.setDataRate(RF24_1MBPS);
  radio.setChannel(108);
  radio.openReadingPipe(0, address);
  radio.startListening();
```

```
}

void loop() {
  if (radio.available()) {
    radio.read(&command, sizeof(command));
    Serial.print("Received: "); Serial.println(command);

    if (command == 'F') {  // Forward
      digitalWrite(M1_IN1, LOW); digitalWrite(M1_IN2, HIGH);
      digitalWrite(M2_IN1, LOW); digitalWrite(M2_IN2, HIGH);
     // digitalWrite(M3_IN1, LOW); digitalWrite(M3_IN2, HIGH);
       //digitalWrite(M4_IN1, LOW); digitalWrite(M4_IN2, HIGH);

    } else if (command == 'B') {  // Backward
      digitalWrite(M1_IN1, HIGH); digitalWrite(M1_IN2, LOW);
      digitalWrite(M2_IN1, HIGH); digitalWrite(M2_IN2, LOW);
      //digitalWrite(M3_IN1, HIGH); digitalWrite(M3_IN2, LOW);
      //digitalWrite(M4_IN1, HIGH); digitalWrite(M4_IN2, LOW);

    } else {  // Stop all motors
      digitalWrite(M1_IN1, LOW); digitalWrite(M1_IN2, LOW);
      digitalWrite(M2_IN1, LOW); digitalWrite(M2_IN2, LOW);
      digitalWrite(M3_IN1, LOW); digitalWrite(M3_IN2, LOW);
      digitalWrite(M4_IN1, LOW); digitalWrite(M4_IN2, LOW);
    }
  }
}
```